

This derivation according to William Pugh – from his skip list Algorithm.

In a skip list, a node is assigned level i or higher with probability p^{i-1} (or p^i depending on 0 – based vs. 1 – based indexing).

0 – based vs. 1 – based indexing → What does it mean ?

1 – based indexing:

- *When William Pugh wrote the theoretical math for the Skip List he started counting the levels at 1.*
- *The Ground Floor : Level 1. Every single node has at least 1 pointer. (0 coin flips needed, Probability = $p^0 = 1$.)*
- *The Next Floor : Level 2. To reach Level 2, a node has to win exactly 1 coin flip. (1 coin flips needed, Probability = p^1 .)*
- *The i^{th} Floor : Level i . To reach Level i , a node has to win exactly $i - 1$ coin flip. ($i - 1$ coin flips needed, Probability = p^{i-1} .)*

Or, we may write :

- *Level L : Required $L - 1$ consecutive coin flips . Probability is p^{L-1} .*

For a list of n elements, the expected number of nodes at level L is given by:

$$***$E[\text{nodes at level } L] = n \cdot p^L$***$$

Analysis suggests that ideally we should start a search at the level `L` where we expect $\frac{1}{p}$ nodes i. e. if $p = \frac{1}{2}$ means it has 2 nodes on left or right (equally likely events).

$$n \cdot p^L = \frac{1}{p}$$

$$\Rightarrow p^L \times p = \frac{1}{n}$$

$$\Rightarrow p^{L+1} = \frac{1}{n}$$

$$\Rightarrow n \times p^{L+1} = 1$$

$$\text{or, roughly } n \times p^L = 1$$

$$\text{or, } p^L = \frac{1}{n}$$

Solving for L:

$$n = \frac{1}{p^L}$$

$$\Rightarrow n = \frac{1}{p^L}$$

$$\Rightarrow n = \left(\frac{1}{p}\right)^L$$

By definition of logarithm [$b^y = x$, then $\log_b(x) = y$]:

$$\Rightarrow L = \log_{\frac{1}{p}}(n)$$

formally William Pugh write it as:

$$\Rightarrow L(n) = \log_{\frac{1}{p}}(n)$$

2. Deriving the expected Maximum Level in a List of `n` Elements and Expected Search Cost through probabilistic analysis:

A **random variable** has a fixed but unpredictable value and a predictable probability distribution and average. If X is a random variable, $\text{Prob}\{X = t\}$ denotes the probability that X equals t and $\text{Prob}\{X > t\}$ denotes the probability that X is greater than t .

Example 1:

Translation: "What is the probability that my die roll (X) exactly equals my target number (t)?"

$\text{Prob}\{X = 6\}$ is exactly $\frac{1}{6}$.

Example 2:

A die has 6 possible outcomes: 1, 2, 3, 4, 5, 6. The question asks for the probability of rolling greater than 3. The winning numbers are 4, 5, and 6. That gives you 3 winning outcomes out of 6 total possibilities.

$$\text{Prob}\{X > 3\} = \frac{3}{6} = \frac{1}{2}.$$

Definition:

Pugh sets up three variables:

- *X : A random variable representing the time it takes to run Algorithm A_X .*
- *Y : A random variable representing the time it takes to run Algorithm A_Y .*
- *t : A specific time limit (for example, 5 milliseconds)*

Definition 1: Probabilistic Equality ($=_{\text{prob}}$)

The formal definition is: $X =_{\text{prob}} Y$ if for all t , $\text{Prob}\{X > t\} = \text{Prob}\{Y > t\}$.

- *The Translation: "Algorithm X and Algorithm Y are probabilistically identical."*
- *What it means: No matter what time limit (t) we pick, both algorithms have the exact same chance of taking longer than that limit.*
- *Example: If there is a 10% chance that Algorithm X takes longer than 5 milliseconds, there is also exactly a 10% chance that Algorithm Y takes longer than 5 milliseconds. If they match for every possible time limit, they are $=_{\text{prob}}$.*

Definition 2: Probabilistic Inequality ($\leq_{\text{prob}}$)

The formal definition is: $X \leq_{\text{prob}} Y$ if for all t , $\text{Prob}\{X > t\} \leq \text{Prob}\{Y > t\}$.

We define $X \leq_{\text{prob}} Y$ to be true if and only if for any value t , the probability that X exceeds t is less than the probability that Y exceeds t .

- *The Translation: "Algorithm X is probabilistically faster than (or equal to) Algorithm Y."*
- *What it means: Pick any time limit (t). The chance that Algorithm X misses that deadline is always smaller than (or equal to) the chance that Algorithm Y misses that deadline.*
- *The Pizza Analogy: Imagine X and Y are pizza delivery places, and t is 30 minutes. If Place X has a 5% chance of taking longer than 30 minutes, and Place Y has a 20% chance of taking longer than 30 minutes, then Place X is probabilistically faster. If this remains true whether t is 10 minutes, 30 minutes, or 2 hours, then $X \leq_{\text{prob}} Y$.*

Let M be a random variable representing the actual maximum level present in a list of n elements.

Probability that a single node exceeds level k : $(p^k)^{1 \text{ node}} = p^k$ `k` due to 1 -
based
indexing

Probability that no node in a list of n nodes exceeds level k :

- Probability that no node in a list exceeds level $k \Rightarrow 1 - p^k$
- And total number of nodes is n , hence: $(1 - p^k)^n$

Bernoulli's Inequality : $(1 - x)^n \geq 1 - nx$

Imagine rolling a 100 sided die. The chance of rolling a 1 is 1%

which means $x = 1\% = \frac{1}{100} = 0.01$.

if we roll it 3 times ($n = 3$), what is the chance we never roll a 1?

- **Exact Math**: $(1 - 0.01)^3 = 0.99^3 \approx 0.970299$
- **Shortcut Math**: $1 - (3 \times 0.01) = 1 - 0.03 = 0.97$

Its visible that 0.970299 is strictly greater than 0.97.

“Pugh uses this rule to simplify his probability equation”

Probability that the maximum level M is greater than k :

$$Prob(M > k) = 1 - (1 - p^k)^n$$

$1 - (1 - p^k)^n$ denotes ``At least one succeeds`` i. e.

Maximum Level $M > \text{Level } k$ and $(1 - p^k)^n$ is the exact chance that the list's height is `k` which is shorter than Maximum height (M), then that the maximum height (M) is greater than k is just 100% minus that failure rate $(1 - p^k)^n$. Here 1 is 100% minus failure rate $(1 - p^k)^n$.

Now, we know ,Bernoulli's Inequality : $(1 - x)^n \geq 1 - nx$ and we have $1 - (1 - p^k)^n$.

Therefore , : $(1 - x)^n \geq 1 - nx$ for $x \in [0, 1]$ (i. e. $0 \leq x \leq 1$)
 $\Rightarrow 1 - (1 - x)^n \leq nx$,

Now plugging this in $\text{Prob}(M > k) = 1 - (1 - p^k)^n$, we get:

$$\text{Prob}(M > k) = 1 - (1 - p^k)^n < n \times p^k \left[\begin{array}{l} \text{As, } (1 - p^k)^n \text{ is strictly greater than} \\ 1 - np^k \text{ i. e., } (1 - p^k)^n > 1 - np^k \\ \text{, the inverted formula} \\ \text{becomes strictly less than:} \\ 1 - (1 - p^k)^n < n \times p^k \end{array} \right]$$

Rewriting $n \times p^k$ using $n = \left(\frac{1}{p}\right)^{L(n)} = p^{-L(n)}$:

$$n \times p^k = \left(\frac{1}{p^k}\right)^{L(n)} = p^{k-L(n)}$$

A binomial distribution is a statistical probability model that counts the number of successes in a fixed number of independent yes – or – no trials.

NB stands for Negative Binomial distribution. Pugh defines $NB(s, p)$ as the number of failures $(1 - p)$ we see while getting our s^{th} success in a sequence of independent Bernoulli trials with success probability p .

Where average:

$$\mathbb{E}[NB(s, p)] = \frac{s(1 - p)}{p}$$

Variance:

$$\text{Var}(NB(s, p)) = \frac{s(1 - p)}{p^2}$$

If ,

The 1 (The Target): This is the s in the formula. It means we are waiting for exactly 1 success. As soon as we get 1 success, the game is over, and we stop counting.

And , $1 - p$ is the probability of success .

Then, it gives rise to $NB(1, 1 - p)$.

where , $s = 1$ and $p = 1 - p$,

$$\text{As, } \mathbb{E}[NB(s, p)] = \frac{s(1 - p)}{p}$$

$$\therefore \mathbb{E}[NB(1, 1 - p)] = \frac{1 \times (1 - (1 - p))}{1 - p} = \frac{p}{1 - p}$$

Where, $1 - (1 - p)$ is the probability of failure .

We know,

- *The i^{th} Floor : Level i . To reach Level i , a node has to win exactly $i - 1$ coin flip. ($i - 1$ coin flips needed, Probability = p^{i-1} .)*

Pugh needs a statistical function that perfectly equals p^i when we ask it for the probability of being greater than i .

To be taller than i , node must reach Level $i + 1$. Level $i + 1$ requires exactly $(i + 1) - 1 = i$ winning coin flips.

Therefore, the probability that a single node is taller than Level i is exactly p^i .

We know,

$$Prob(M > k) < n \times p^k$$

Note:

- *M (The Reality): This is the actual, physical maximum height that our skip list ended up growing to.*
- *$L(n)$ (The Target): This is the expected, ideal baseline search level.*
Expected nodes at level $L(n)$ or higher = $\frac{1}{p}$.

If there are n elements and each node has probability p of rising from level i to level $i + 1$, then the expected number of nodes with level at least L is

$$np^{L-1} [\mathbb{E}[N_{\geq L}] = np^{L-1}]. \text{ Pugh defines } L(n) = \log_{\frac{1}{p}} n$$

as the level where the expected number of nodes at level $L(n)$ or higher is $\frac{1}{p}$. It is the ideal starting level for a search.

- ***k (The Test Subject): This is an arbitrary variable used to test boundaries.***

$$i = k - L(n).$$

By ,Rearranging this ,we get : $k = L(n) + i$.

Now ,we want to test a level `k` that sits exactly `i` floors above our expected target $L(n)$.

So,we set our test level to $k = L(n) + i$.

$$\begin{aligned} \text{Prob}(M > k) &< n \times p^k \\ \Rightarrow \text{Prob}(M > L(n) + i) &< n \times p^{L(n)+i} \end{aligned}$$

$$\text{Now, } n \times p^{L(n)+i} = n \times p^{L(n)} \times p^i$$

$$\text{But , } L(n) = \log_{\frac{1}{p}}(n)$$

$$\text{So, } p^{L(n)} = p^{\log_{\frac{1}{p}} n}$$

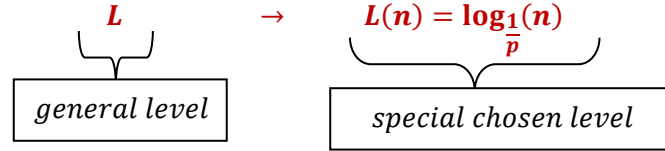
We know, $\left(\frac{1}{p}\right)^L = n$ and we know, $L = \log_{\frac{1}{p}}(n)$, Therefore: $\left(\frac{1}{p}\right)^{\log_{\frac{1}{p}}(n)} = n$,

since $\frac{1}{p} = p^{-1}$, we have $p^{-\log_{\frac{1}{p}}(n)} = n$, Taking reciprocal , we get : $p^{\log_{\frac{1}{p}}(n)} = n^{-1} = \frac{1}{n}$.

$$\therefore p^{L(n)} = p^{\log_{\frac{1}{p}} n} = \frac{1}{n}$$

$$\text{Now,plugging it in: } n \times p^{L(n)} \times p^i = n \times \frac{1}{n} \times p^i = p^i.$$

Also, from here we can also prove that : $\mathbb{E}[N_{\geq L}] = np^{L-1} = n \times p^L \times p^{-1}$
 $= n \times p^L \times p^{-1} = n \times p^L \times \frac{1}{p}$, and we know $L(\text{general level}) = \log_{\frac{1}{p}}(n)$,
then Pugh names that particular value as $L(n)$. We can think of it as :



i. e., When $L = L(n)$, then: $n \times p^{L(n)} \times \frac{1}{p} = n \times p^{\log_{\frac{1}{p}}(n)} \times \frac{1}{p} = n \times \frac{1}{n} \times \frac{1}{p} = \frac{1}{p}$.

$$\therefore \mathbb{E}[N_{\geq L}] = \frac{1}{p},$$

Now , coming back:

$$\begin{aligned} \text{Prob}(M > k) &< n \times p^k \\ \Rightarrow \text{Prob}(M > L(n) + i) &< n \times p^{L(n)+i} \\ \Rightarrow \text{Prob}(M > L(n) + i) &< n \times p^{L(n)} \times p^i \\ \Rightarrow \text{Prob}(M > L(n) + i) &< p^i \end{aligned}$$

Pugh also observes :

$$\text{Prob}(NB(1, 1 - p) + 1 > i) = p^i$$

Where, $NB(1, 1 - p) + 1$ represents the few incredibly lucky nodes that randomly flipped ``Heads`` over and over again, growing past the $L(n)$ target.

More precisely, $NB(1, 1 - p) + 1$ represents the random extra number of levels by which the maximum node level can extend beyond $L(n)$, corresponding to an unusually long sequence of successful promotions (the node keeps getting promoted with probability p).

We know,

$$\begin{aligned}\mathbb{E}[NB(1, 1 - p)] &= \frac{p}{1 - p} \\ \therefore \mathbb{E}[L(n) + NB(1, 1 - p)] &= L(n) + \frac{p}{1 - p} \\ \therefore \mathbb{E}[L(n) + NB(1, 1 - p) + 1] &= L(n) + \frac{p}{1 - p} + 1 \\ &= L(n) + \frac{p + 1 - p}{1 - p} \\ &= L(n) + \frac{1}{1 - p}\end{aligned}$$

$$\text{Hence , } E[M] \leq L(n) + \frac{1}{1 - p}$$

Lets rearrange:

$$\begin{aligned}Prob(M > k) &< n \times p^k \\ \Rightarrow Prob(M > L(n) + i) &< n \times p^{L(n)+i} \\ \Rightarrow Prob(M > L(n) + i) &< n \times p^{L(n)} \times p^i \\ \Rightarrow Prob(M > L(n) + i) &< p^i\end{aligned}$$

And ,

$$Prob(NB(1, 1 - p) + 1 > i) = p^i$$

Hence,

$$Prob(M > L(n) + i) < Prob(NB(1, 1 - p) + 1 > i)$$

This allows Pugh to establish a formal probabistic upper bound for M:

$$\text{Therefore , } M \leq_{prob} L(n) + NB(1, 1 - p) + 1$$

The symbol $\leq_{\text{prob}}$ means is probabilistically less than or equal to. Pugh is establishing a rock – solid upper limit on how tall the skip list (M) can possibly get.

Where,

- M : The actual, physical maximum level of the tallest node in our skip list of n elements.
- $\leq_{\text{prob}}$: "Probabilistically less than or equal to." It means, "I guarantee the height will almost never exceed the right side of this equation".
- $L(n)$: The Expected Target. This is the base height we expect a list of n elements to reach.
- Where, $NB(1, 1 - p) + 1$ represents the few incredibly lucky nodes that randomly flipped "Heads" over and over again, growing past the $L(n)$ target.

Translated into plain English, this formula means:

"The maximum height of the entire skip list (M) will safely be less than or equal to our ideal target height ($L(n)$), plus a tiny, rapidly shrinking statistical streak of extra coin flips ($NB(1, 1 - p) + 1$)."

Now,

$$\text{Expected value of } M \text{ or Average of } M \text{ i.e. } E[M] \leq L(n) + \frac{1}{1 - p}$$

Definition (binomial distributions – $B(t, p)$). Let t be a non – negative integer and p be a probability. The term $B(t, p)$ denotes a random variable equal to the number of successes seen in a series of t independent random trials where the probability of a success in a trial is p .

Where average:

$$\mathbb{E}[B(t, p)] = t \times p$$

Variance:

$$\text{Var}(B(t, p)) = t \times p(1 - p)$$

Where,

- $t \rightarrow$ The trials or the total number of attempts we get to flip coins.
- $p \rightarrow$ The probability of success on any single attempt.
- $1 - p \rightarrow$ The probability of failure.

$$\text{We are at: } M \leq_{\text{prob}} L(n) + NB(1, 1 - p) + 1$$

So the maximum level might extend to:

$$L(n) + NB(1, 1 - p) + 1$$

$$\text{Expected value of } M \text{ or Average of } M \text{ i.e. } E[M] \leq L(n) + \frac{1}{1 - p}$$

- $L(n) - 1 \rightarrow$ Upward movements from Level 1 to $L(n)$.
- $NB(L(n) - 1, p) \rightarrow$ Leftward movements needed while finding those $L(n) - 1$ upward moves.

$$\therefore \mathbb{E}[(L(n) - 1 + NB(L(n) - 1, p))], \text{ we know } \mathbb{E}[NB(s, p)] = \frac{s(1 - p)}{p}$$

$$\therefore NB((L(n) - 1, p)) = \frac{(L(n) - 1)(1 - p)}{p}$$

$$\Rightarrow \mathbb{E}[(L(n) - 1 + NB(L(n) - 1, p))] = L(n) - 1 + \frac{(L(n) - 1)(1 - p)}{p}$$

$$\begin{aligned}
&= L(n) - 1 \left(1 + \frac{(1-p)}{p} \right) \\
&= L(n) - 1 \left(\frac{p + 1 - p}{p} \right) \\
&= L(n) - 1 \left(\frac{1}{p} \right) \\
&= \frac{L(n) - 1}{p}
\end{aligned}$$

- $B\left(n, \frac{1}{np}\right) \rightarrow$ *Leftward movements remaining once at $L(n)$, bounded by the number of nodes with level $\geq L(n)$.*

$$p^{L(n)-1} = p^{L(n)} \times p^{-1} = p^{\frac{\log_1(n)}{p}} \times \frac{1}{p} = \frac{1}{n} \times \frac{1}{p} = \frac{1}{np}, \text{ Therefore , } B\left(n, \frac{1}{np}\right).$$

$$\therefore \mathbb{E} \left[B\left(n, \frac{1}{np}\right) \right] = n \times \frac{1}{np} = \frac{1}{p} [As, \mathbb{E}[B(t, p)] = t \times p]$$

- $NB(1, 1-p) + 1 \rightarrow$ *Extra upward movements from $L(n)$ to the unusually high maximum level.*

Where pugh defines M as the maximum level and establishes:

$$M \leq_{prob} L(n) + NB(1, 1-p) + 1$$

Therefore ,once we have already reached $L(n)$, the additional upward distance is:

$$[L(n) + NB(1, 1-p) + 1] - L(n) = NB(1, 1-p) + 1.$$

$$As, \mathbb{E}[NB(s, p)] = \frac{s(1-p)}{p}$$

$$\therefore \mathbb{E}[NB(1, 1-p)] = \frac{1 \times (1 - (1-p))}{1-p} = \frac{p}{1-p}$$

$$\therefore \mathbb{E}[NB(1, 1-p) + 1] = \frac{p}{1-p} + 1 = \frac{p + 1 - p}{1-p} = \frac{1}{1-p}$$

$\therefore$ Cost of the entire search is $:=$

Total cost to climb out of a list of `n` elements

$$\leq \text{prob}(L(n) - 1 + NB(L(n) - 1, p))$$

reach $L(n)$

$$+ B\left(n, \frac{1}{np}\right)$$

left movements at or above $L(n)$

$$+ NB(1, 1-p) + 1$$

climb above $L(n)$

This is what Pugh means by a probabilistic upper bound on the total length of the search path.

Then taking expectations gives :

$\mathbb{E}[\text{search cost}]$

$$= \mathbb{E}[(L(n) - 1 + NB(L(n) - 1, p))] + \mathbb{E}\left[B\left(n, \frac{1}{np}\right)\right] + \mathbb{E}[NB(1, 1-p) + 1]$$

$$= \frac{L(n) - 1}{p} + \frac{1}{p} + \frac{1}{1-p}$$

$$\begin{aligned}
&= \frac{L(n)}{p} + \frac{1}{1-p} \\
&= \frac{L(n)}{p} + \frac{1}{1-p} \\
&= \frac{\log_{\frac{1}{p}}(n)}{p} + \frac{1}{1-p}
\end{aligned}$$

Plugging $p = 0.5$, we get:

$$\begin{aligned}
&= \frac{\log_{\frac{1}{0.5}}(n)}{0.5} + \frac{1}{1-0.5} \\
&= \frac{\log_{\frac{1}{2}}(n)}{\frac{1}{2}} + \frac{1}{1-\frac{1}{2}} \\
&= 2 \times \log_2 n + 2 \\
&= O(2 \times \log_2 n + 2) \\
&= O(\log_2 n)
\end{aligned}$$

Or we can say as $0 < p < 1$ since p is a probability, hence $\frac{1}{1-p}$ and $\frac{1}{p}$ are all constants, hence, $\mathbb{E}[\text{search cost}] = O(\log n)$.
